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Investigation of Fuel and Load Flexibility in a Siemens Gas Turbine-600/700/800 Burner Under Atmospheric Pressure Conditions Using High-Speed Hydroxyl-PLIF and Hydroxyl Radical Chemiluminescence Imaging

Fuel and load flexibility have been increasingly important features of industrial gas turbines in order to meet the demand for increased utilization of renewable fuels and to provide a way to balance the grid fluctuations due to the unsteady supply of wind and solar power. Experimental investigations were performed using a standard third-generation dry low emission (DLE) burner under atmospheric pressure conditions to study the effect of central and pilot fuel addition, load variations, and hydrogen (H_2) enrichment in a natural gas (NG) flame. High-speed kHz planar laser-induced fluorescence (PLIF) of OH radicals and imaging of OH chemiluminescence were employed to investigate the flame stabilization, flame turbulence interactions, and flame dynamics. Along with the optical measurements, combustion emissions were also recorded to observe the effect of changing operating conditions on NO_X level. The burner is used in Siemens industrial gas turbines SGT-600, SGT-700, and SGT-800 with minor hardware differences and the study thus is a step to characterize fuel and load flexibility for these turbines. Without pilot and central fuel injections in the current burner configuration, the main flame is stabilized creating a central recirculation zone (CRZ). Addition of the pilot fuel strengthens the outer recirculation zone (ORZ) and moves the flame anchoring position slightly downstream, whereas the flame moves upstream without affecting the ORZ when central fuel injection is added. The flame was investigated utilizing H_2/NG fuel mixtures where the H_2 amount was changed from 0 to 100%. The results show that the characteristics of the flames are clearly affected by the addition of H_2 and by the load variations. The flame becomes more compact, the anchoring position moves closer to the burner exit and the OH signal distribution becomes more distinct for H_2 addition due to increased reaction rate, diffusivity, and laminar burning velocity. Changing the load from part to base, similar trends were observed in the flame behavior but in this case due to the higher heat release because of increased turbulence intensity. [DOI: 10.1115/1.4049499]

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Introduction

Increasing demand for utilizing a wide range of alternative and renewable fuels makes it imperative to consider operational flexibility, efficiency, fuel flexibility, and emissions when designing modern industrial gas turbine engines. One key challenge is to achieve high efficiency combined with low emissions for different liquid and gaseous fuels at load variations. Currently, stationary gas turbine engines are designed to operate under lean premixed conditions in order to reduce thermal NO_X production [1,2] by lowering the flame temperature. However, this approach can lead to undesirable instabilities, which may cause local flame extinction, thermoacoustic oscillations, and poor combustion efficiency [3]. To design an industrial gas turbine, eliminating such instabilities is a significant challenge. The inclusion of a diffusion pilot flame or a prechamber combustor can prevent the occurrence of such instabilities [4,5]. A range of alternative fuels, such as residual product gases from refineries, chemical plants, and petroleum industries, particularly renewable fuels derived from biomass, have been proposed for use in gas turbines due to a substantial cost advantage, changes in the supply and availability of traditional fuel, and minimization of the adverse effects on the environment. These fuels commonly contain a significant fraction of hydrogen (H_2) and hydrocarbons (HC). H_2, a carbon-free energy carrier, can also be produced from renewable energy sources, such as solar and wind power, via direct thermal conversion or by electrolysis [6,7]. Several combustion relevant properties of H_2, e.g., high diffusivity, wide flammability range, high laminar and turbulent flame speed, are significantly different from those of a conventional HC fuel and can affect the operational limits, flame stabilization, and exhaust gas emissions [8–13]. The enrichment of H_2 increases the range of flame stability and lowers the lean blowout (LBO) limit [8], while it makes the flame more sensitive to flashback by enhancing the laminar and turbulent flame speed [11]. H_2 enrichment can also affect the flame stabilization by altering the flame shape and position [12]. The DNS simulation by Hawkes and Chen [14] shows that the flame surface area can be increased due to the higher burning rate per unit area. An increase in NO_X and dynamic modes of the combustor were observed in the study by Lam and Parsania [15].

The primary objective of the current study is to experimentally investigate the effects of central and pilot fuel addition, load variations and H_2 enrichment in the natural gas (NG) flame in a Siemens standard third generation dry low emission (DLE) burner under atmospheric conditions. Although high pressure is a required condition for a real gas turbine, relevant phenomena, e.g., flame turbulence interactions and flame stabilization, can also be investigated at atmospheric pressure for fundamental understanding. In addition, it is much less challenging to evaluate designs and combustion conditions using advanced measurement tools, such as laser techniques, at atmospheric pressure in order to explain phenomena detected but not fully understood at higher pressure. High-speed OH planar laser-induced fluorescence (PLIF) and OH chemiluminescence were employed in the current investigations for temporally resolved studies of flame structure, flame stabilization, and dynamics. The OH chemiluminescence images are useful for acquiring information about the overall size and shape of the flame while the OH-PLIF images indicate the reaction zone and postflame regions. Finally, NO_X emissions were recorded to observe the effect of changing operating conditions.

This burner is used in the industrial gas turbines SGT-600 (25 MW), SGT-700 (33 MW), and SGT-800 (62 MW) with minor hardware differences. The burner design principle is well-proven since its early development history [16] and improvements are constantly implemented [17,18]. This type of burner has been thoroughly investigated using different methods such as CFD and measurements using water rig for investigating the fuel-air mixing, atmospheric combustion rig tests for fundamental investigations with high optical access, high-pressure rig tests to include the pressure effect and engine tests for complete engine conditions and the effect of neighboring burners. More details of some of the performed investigations are presented in Refs. [17–22].

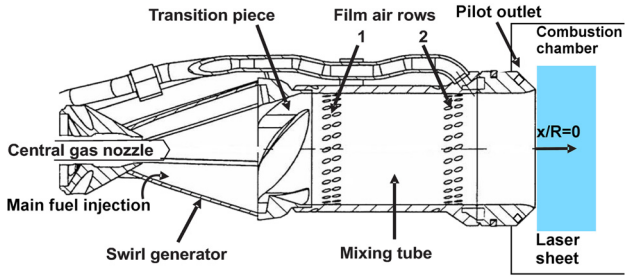

Fig. 1 Schematic of the Siemens third generation DLE burner from Ref. [19] with laser beam path and combustion chamber added

Experimental Setup

Burner and Atmospheric Combustion Test Rig. Measurements were performed in the Siemens standard third generation DLE burner and its schematic is presented in Fig. 1. The airflow enters a swirl generator consisting of four quarter cones shifted with respect to each other to create a swirling flow. The swirl generator is connected to a circular mixing tube with a secondary air inlet by means of film air rows. After main fuel injection in the swirl generator, the swirling air/fuel mixture passes through the mixing tube, after which it enters the combustion chamber where it expands in a radial direction and ignites as it mixes with hot recirculated combustion gases. The burner configuration also allows the injection of additional fuel through pilot outlets, which are equidistantly distributed at the circumference around the burner tip, and through the central gas nozzle shown in Fig. 1.

The DLE burner was mounted in an atmospheric test rig also used in previous studies [17,18,20]. The experimental setup in the atmospheric rig is shown in Fig. 2. The combustion chamber located after the burner exit consists of a convectively cooled liner. The circular cross section of the combustion chamber has a similar expansion ratio as in the real gas turbine annular combustors; however, the initial section just after the burner exit has square cross section with smoothed corners to improve measurement quality. Access for optical diagnostics in the flame region is provided through quartz windows on three sides of the liner. Before entering the plenum, the combustion air was preheated up to 693 K. The fuel flow rates to the main, pilot, and central gas injectors were controlled by means of Coriolis mass flow controllers.

Operating Conditions. The experiments were performed at atmospheric pressure conditions while inlet air temperature, burner pressure drop, and flame temperature were downscaled and altered to engine conditions by matching the reference velocity in the mixing tube (based on the mass flow) and the equivalence ratio. Three parameters were varied to study their effects on combustion: Central and pilot fuel addition, load variation, and H_2 addition. At the central fuel to total fuel ratio (CFR) = 2% setting, 2% of the total fuel amount was injected through the central gas nozzle shown at the left of Fig. 1, while at the pilot fuel to total fuel ratio (PFR) = 3% setting, 3% of the total fuel amount was guided to the pilot outlet so that small diffusion flames were introduced around the edge of the burner tip. Finally, at PFR = 0, CFR = 0 setting, only main fuel was injected in the swirl generator without supplying fuel through the pilot and central injectors. When changing the fuel route, the bulk airflow was kept constant. In addition to the effect of guiding fuel to different paths, the effects of load variations, from part load (about 80%) to base load, were studied by varying the mass flow rate of the gas mixture. Both the mass flow of the air and the mass flow of the fuel

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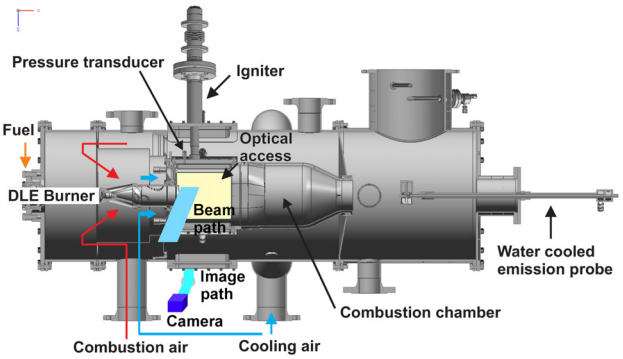

Fig. 2 Schematic of atmospheric combustion test rig assembly from Ref. [19] with laser beam path and image detection system added

were raised by a factor 1.2 to achieve the base load condition compared to the part load condition. The inlet air temperature was also increased by about 20 K at the base load condition compared to the part load. The same adiabatic flame temperature was used for all cases including the part load case. The reason for this is that the twin shaft engines SGT-600 and SGT-700 have combustor air bypass system in order to keep similar flame temperature despite reduced turbine inlet temperature. For the single shaft engine SGT-800, this is also relevant since the turbine inlet temperature is barely affected down to about 75% load using normal settings. The flame was also investigated using pure NG as well as pure H_2 fuel. In addition to the pure H_2 combustion case, H_2 enrichment where the NG fuel was enriched with 30 vol %, 60 vol %, and 80 vol % of H_2 was studied at the PFR = 3% setting. Here, the airflow was kept constant, while the mass flow rate of the fuel was adjusted at each H_2 concentration to achieve a constant global adiabatic flame temperature, which is a typical control strategy for practical gas turbines. Air-to-fuel ratio (AFR) at each flame condition is normalized by the air-to-fuel ratio at the base load (NG) case with PFR = 0, CFR = 0 (AFR0) and the normalized values are tabulated in Table 1.

Diagnostics

As optical measurement techniques, high-speed OH-PLIF, and OH chemiluminescence imaging were employed for investigating the flame at the burner exit.

OH-Planar Laser-Induced Fluorescence

High-speed (kHz) OH-PLIF measurements were performed by excitation of the $A^2\Sigma^+ - X^2\Pi$ (1–0) band of the OH radical. A frequency-doubled (532 nm) pulsed solid-state diode-pumped Nd:YAG laser (Edgewave HD40I-E) was used to pump a tunable dye laser (Sirah Credo) operated with Rhodamine 590 dye. The output of the dye laser (~566 nm) was frequency-doubled and the laser wavelength was tuned to the temperature-insensitive $Q_1(8)$ transition at 283.55 nm. An approximately 50 mm wide laser sheet was formed using an f = −40 mm cylindrical lens and an f = +500 mm spherical lens and passed along the midline axis of the burner exit. The laser pulse energy varied between 1.75 mJ and 1.9 mJ during the measurements. The resulting OH fluorescence signal at wavelength ~310 nm was collected perpendicular to the laser sheet with a high-speed CMOS camera (Fastcam SA5) in combination with a two-stage image intensifier (HiCATT). The detector system was fitted with a UV sensitive lens (B. Halle, f#2, f = 100 mm). To suppress interference occurring from scattered laser light and background radiation, a single-band bandpass filter (315/15 nm BrightLine®) was mounted in front of the lens. The detector was set to a time gate of 40 ns to further limit background contributions from flame chemiluminescence. For each of the flame conditions, 2728 single-shot images (using max. buffer memory of the camera) were collected at 2 kHz frequency. For compensation in the average spatial profile of the laser sheet, a cuvette with a weak dye solution was used. The scaling factor of the acquired images was 0.15 mm/pixel.

Flame Chemiluminescence. The same detector system was used to acquire line-of-sight integrated chemiluminescence images of excited OH radicals (OH*) at wavelength 310 nm. The camera was equipped with the same UV sensitive lens (B. Halle, f#2, f = 100 mm) and a bandpass filter centered at 310 nm. For each flame condition, 2728 single-shot images were collected at a frequency of 2 kHz and with an exposure time of 20 μs.

Results and Discussion

Effect of Central Fuel to Total Fuel Ratio and Pilot Fuel to Total Fuel Ratio on the Main Flame. In Fig. 3, OH-PLIF images show the effects of the pilot and central fuel addition on the NG/air flame structure at a base load condition. OH-PLIF measurements were carried out to provide qualitative information of OH radical distributions in the symmetry plane through the center of the burner exit and identify the reaction zone and the postflame region. The high-intensity region indicates OH radicals that are formed in the flame front or have been convected a relatively short distance [23]. The coordinates of the images are normalized by the burner exit inner radius (R) and the flow direction is from left to right. In Figure 3, representative single shots and evaluated gradients of OH are shown in the first two rows, while probability density maps (PDMs) of the OH gradients and single shots are shown in the last two rows. The strongest gradients are good markers to indicate the approximate position of the flame front [24,25], while weaker gradients correspond to boundary layers between hot and cold gases. For each case, the PDM images were calculated by binarizing and later normalizing 2728 single shots as well as corresponding gradient images, to present the spatial distributions of OH radicals and flame fronts. A local thresholding scheme was implemented to identify the boundaries of the reaction layers and to form binarized images. For each column of pixels, 10% of the local maximum (in that column) was selected as a threshold for the signal-to-background ratio. The width of PDMs of the gradients indicates the flame front fluctuations.

Previous experimental and LES investigations [18,21] of this burner configuration showed that the main flame is stabilized downstream of the burner exit due to a central recirculation zone (CRZ) induced by the vortex breakdown process. The swirling air generated by the swirler and fuel introduced through injectors in the swirler can achieve good mixing after passing through the mixing tube and then discharge into the combustor through the burner nozzle, where it expands radially by the imposed swirl [26]. The main flame is anchored at the inner shear layer (ISL) formed by the CRZ, which brings heat and radicals from the main flame toward the burner exit and initiates reactions with the fresh fuel/air mixture. A sharp temperature and OH gradient indicate the flame front in the vicinity of the reaction zone. In the current results presented in Fig. 3, only the early portion of the NG/air flame, covered by the width of the laser sheet, is shown for three CFR and PFR settings: (a) PFR = 0, CFR = 0, (b) PFR = 3%, CFR = 0, (c) PFR = 3%, CFR = 2%. The conical shape of the OH distribution with high intensity, found at the ISL of the swirling air/fuel mixture, indicates the leading edge of the CRZ, whereas low OH signal found along the outer shear layer (OSL) indicates the outer recirculation zone (ORZ). The flame exhibits repeatedly large axial displacements with superimposed local fluctuations due to various scales of turbulence. An arbitrary cycle of the main flame fluctuation is shown in a time sequence in Fig. 4. In the sequence, the downstream propagation of the flame is clearly observed up to 0.2 ms, after which the flame is almost out of the field of view. Then, it is found to be pulled back again in the

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Table 1 Fuel gas composition and normalized AFR values for different flame conditions

Operating condition	Fuel gas composition (Vol %) NG	H_2	AFR/AFR0 AFR/AFR0	Fuel splitting Main fuel (%)	Pilot fuel [%]	Central fuel (%)
Base load (NG): PFR = 0, CFR = 0	100	0	1	100	0	0
Base load (NG): PFR = 3%, CFR = 0	100	0	1	97	3	0
Base load (NG): PFR = 3%, CFR = 2%	100	0	1	95	3	2
Part load (NG): PFR = 3%, CFR = 2%	100	0	1	95	3	2
Base load (30% H_2): PFR = 3%, CFR = 0	70	30	1.06	97	3	0
Base load (60% H_2): PFR = 3%, CFR = 0	40	60	1.21	97	3	0
Base load (80% H_2): PFR = 3%, CFR = 0	20	80	1.45	97	3	0
Base load (100% H_2): PFR = 0, CFR = 0	0	100	2.36	100	0	0

The adiabatic flame temperature was the same for all flame conditions.

frame from 0.5 ms and to propagate toward the burner exit. These repeated displacements of the flame explain why the probability values, indicating the probability for occurrence of the flame at a specific position, are small in the PDM images shown in Fig. 3IV.

Twelve pilot fuel injectors are arranged equidistantly at the circumference of the burner exit. Since the pilot fuel and co-airflow can achieve very limited mixing, the combustion occurs in a diffusion manner while interacting with the main flame. This pilot flame can be very useful to improve the LBO limit or flame stability close to the LBO conditions. Considerable effects on the flame structure were observed due to the addition of this pilot fuel. At

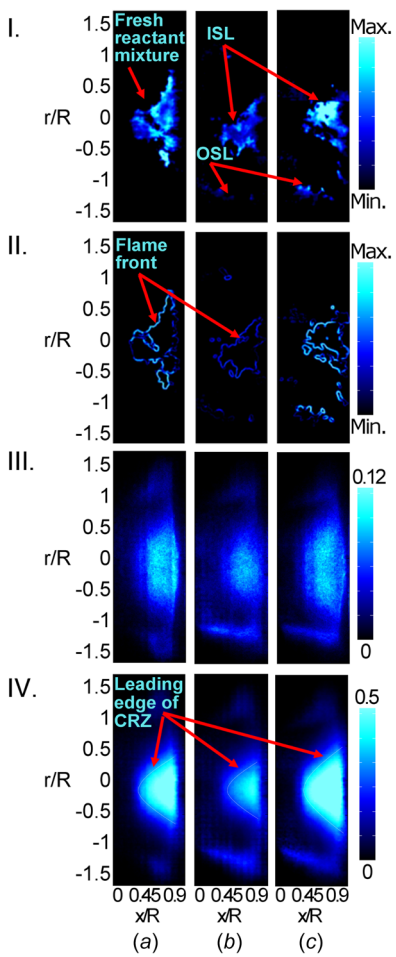

Fig. 3 OH-PLIF results of NG/air flame at base load conditions with three CFR and PFR settings: (*a*) PFR = 0, CFR = 0, (*b*) PFR = 3%, CFR = 0, and (*c*) PFR = 3%, CFR = 2%. I. Single-shots of OH-PLIF; II. Gradients of OH; III. PDMs of OH gradient; IV. PDMs of single-shots.

PFR = 3%, CFR = 0 setting, the pilot fuel intensified reactions at the ORZ, and the flame was found to be anchored attaching to the burner tip, which is consistent with the observation reported in a previous study of an older version of the burner [18]. However, the spread of the OH distribution at the ORZ was much higher and the flame attachment was much stronger in the previous burner [18] as compared to the current one. When adding the pilot fuel, the flame was also observed to be stabilized slightly downstream in the combustor as compared to the base case when neither pilot nor central fuel was added. In contrast, at the PFR = 3%, CFR = 2% setting, the stabilization of the flame moved upstream again for the additional injection of the central fuel (cf. Fig. 3(*c*) III/IV). The addition of central fuel enhances the CRZ, without affecting the ORZ, by increasing the fuel quantity or the local equivalence ratio in the central part of the fuel/air mixing profile upstream the main flame. For all cases with PFR = 3%, the enhancement of the ORZ was observed only in the lower part of the flame images. This effect is not related to the absorption of the laser along the beam path through the flame, which can be confirmed by the symmetric distribution of OH downstream. However, one reason could be an asymmetry in the flow distribution of the cooling air in the thin channels around the burner. It could create areas in which the local additional air would quench the neighboring pilot flames. This is more prominent when the PFR values are below 3%. Another reason could be the difference in the behavior of active pilots while the strength of the pilot flame varies over time with significantly low frequencies.

The PDMs of gradients indicate that for all cases, the flame front at the ISL fluctuates more in both axial and radial direction as compared to that at the OSL. Figure 5 presents the probability distributions of OH radicals along the radius at five different axial positions. The probability distributions also indicate that the ORZ is not symmetric at different axial positions. The axial distributions at $r/R < -0.75$ and $x/R < 0.45$ verify again the attachment of flame to the burner tip due to the addition of pilot fuel, while the radial distributions at $-1 < r/R < 1$ at different axial positions indicate that the flame moves downstream with pilot fuel injection and moves upstream with central fuel addition.

In Fig. 6, time-averaged OH chemiluminescence images, indicating the shape and position of flame, confirm the flame again to be stabilized downstream in comparison to the base case for adding 3% pilot fuel, while it moves upstream for an additional injection of 2% central fuel.

Effect of Load Variation. Figure 7 presents the effects of load variations on the NG/air flame structure. Comparison of the OH distribution at two conditions, part load and base load at PFR = 3%, CFR = 2%, indicates that the CRZ became more intense and compact for increasing the load. With increased load, keeping ϕ constant, the turbulence level and turbulent flame speed increase with the Reynolds number. The observed movement of the CRZ toward the burner exit at the base load condition indicates an increase in flame speed. At the base load, the ORZ also

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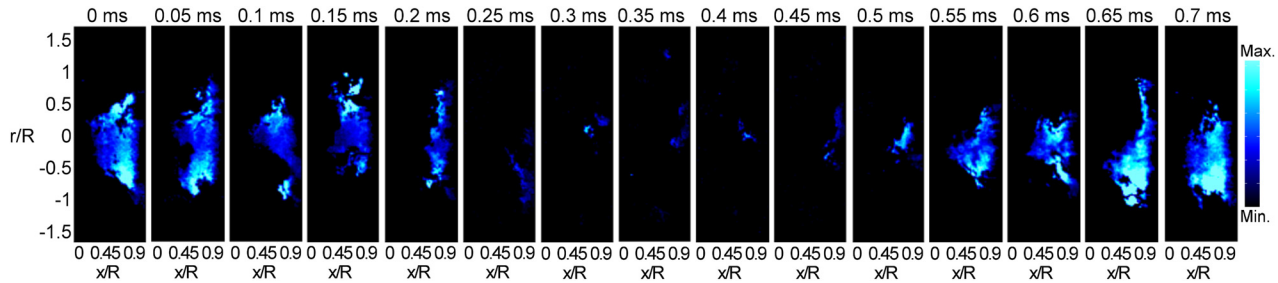

Fig. 4 The displacement of the flame is shown by OH distribution in time sequence. Base load with PFR = 0 and CFR = 0 setting.

became thinner (cf. Fig. 7(*b*) III) due to the faster burning rate of the pilot fuel while interacting with the relatively closer located main flame as compared to the part load case. Furthermore, the PDMs of gradients indicate that the flame front at the OSL fluctuates more in both axial and radial directions at the part load condition as compared to the base load. Like the base load case, the axial displacements of the flame including fluctuations were also observed for the part load case.

Similar to the PLIF images, time-averaged OH chemiluminescence images, shown in Fig. 8, indicate higher intensity for the flame at base load, which is attributed to a higher production rate of OH radicals from chain branching and chain propagation reactions [27]. The upstream stabilization of the base load flame is also observed in the chemiluminescence images.

Effect of H_2 Enrichment. Figure 9 presents the effects of changing fuel, from pure NG to pure H_2, on the flame structure at base load conditions with PFR = 0 and CFR = 0. For a pure H_2 case, the characteristics of the flame are significantly affected by the faster reaction rate, higher diffusivity, and higher burning velocity of H_2. Although the flame still follows the same stabilization mechanism as for pure NG, apparent differences are observed in the flame shape and anchoring position for the pure H_2 fuel. The higher reactivity and flame speed associated with H_2 cause the flame to be shorter and more compact. The single-shot and PDM images indicate the flame to be stabilized quite close to the burner exit. Despite that the OH signal at both the CRZ and ORZ was quite intense, the flame was not observed to be attached to the burner tip.

The PDMs of gradients suggest that the flame front at the ISL fluctuates significantly less in both axial and radial direction for pure H_2 flame as compared to the NG flame. Intensity fluctuations were also observed in the flame, which is shown in Fig. 10 by the variation of the OH signal intensity in a time sequence. Radial distributions of probability for OH at different axial positions for NG and H_2-enriched NG flames are shown in Fig. 11. It also verifies that the pure H_2 flame at PFR = 0 and CFR = 0 setting is stabilized by both CRZ and ORZ but is not attached to the burner tip.

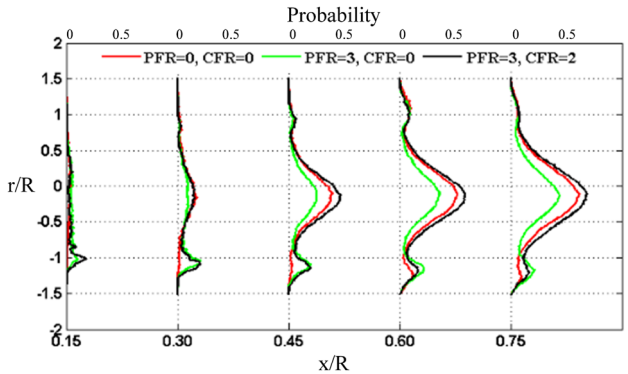

Fig. 5 Radial probability distributions of OH at five different axial positions obtained from OH-PLIF images for PFR = 0–3% and CFR = 0–2% settings

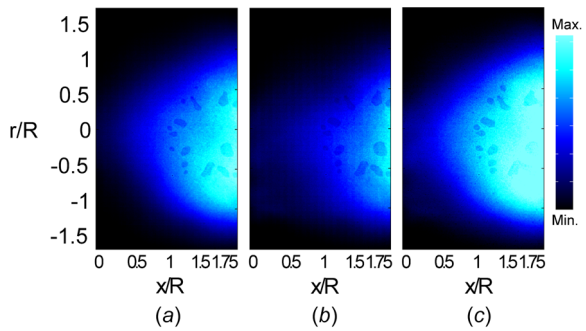

Fig. 6 Time-averaged OH chemiluminescence at three CFR and PFR settings: (*a*) PFR = 0, CFR = 0, (*b*) PFR = 3%, CFR = 0, and (*c*) PFR = 3%, CFR = 2% settings

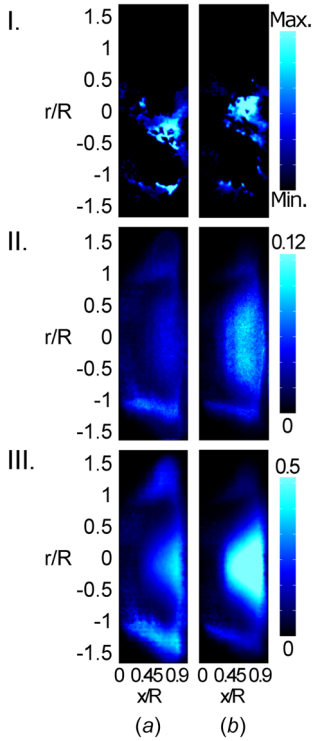

Fig. 7 OH-PLIF results of NG/air flame at different loads: (*a*) part load PFR = 3%, CFR = 2%, (*b*) base load PFR = 3%, CFR = 2%. I. Single-shots of OH-PLIF; II. PDMs of OH gradient; III. PDMs of single-shots.

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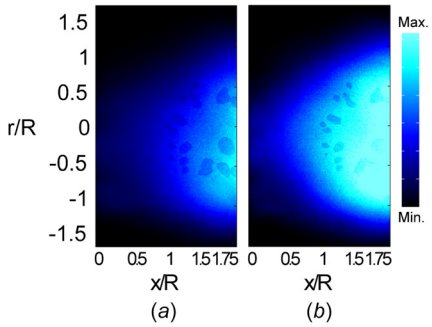

Fig. 8 Time-averaged OH chemiluminescence at different loads: (*a*) part load at PFR = 3%, CFR = 2% and (*b*) base load at PFR = 3%, CFR = 2%

The time-averaged OH chemiluminescence in Fig. 12 and single-shot OH-PLIF images in Fig. 9 show stronger OH* and OH LIF signal for the H_2 flame as compared to the NG flame, which is attributed to the higher production rate of OH for burning the more reactive H_2 fuel promoting radical formation via chain branching and chain propagation reactions [27]. The time-averaged OH chemiluminescence images also indicate that the H_2 flame is more compact and stabilized quite close to the burner exit.

At base load condition with PFR = 3% and CFR = 0 setting, the influence of H_2 enrichment (up to 80 vol %) on NG flame structure is shown in Fig. 13. The ORZ became stronger due to addition of pilot fuel and therefore, the flame was observed (cf. Fig. 10) to be anchored attaching to the burner tip. Besides the pilot fuel effects, the stabilization of the H_2-enriched NG flame followed the same mechanism as the pure H_2 flame. The flame became more compact and shorter as well as anchored closer to the burner exit with H_2 enrichment. The fact that a variety of flames (pure NG flames, H_2-enriched flames, and pure H_2 flames) could be stabilized in this burner demonstrates the improvement and development in burner design toward fuel flexibility. In a previous study [17] with a similar burner at PFR = 0, CFR = 0, only NG flames enriched up to 80 vol % H_2 could be stabilized. Also, the flame at 80 vol % H_2 was stabilized with help of a flame attachment to the burner tip, which can cause unwanted increase of the burner surface temperature and, in worst cases, a triggering of an upstream flame propagation. Thus, the current burner design with no flame attachment to the burner tip (see Figs. 9 and 13) is more optimized for fuel flexibility operation than the one used in Ref. [17].

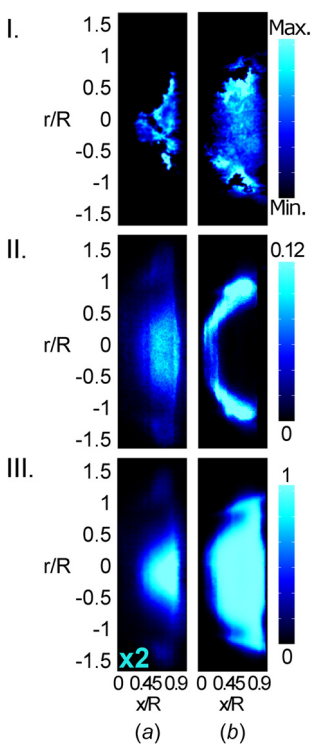

Fig. 9 OH-PLIF results of (*a*) NG/air and (*b*) H_2/air flame at base load with PFR = 0 and CFR = 0. I. Single-shots of OH-PLIF; II. PDMs of OH gradient; III. PDMs of single-shots.

Flame Fluctuations. For all investigated cases, combustion fluctuations were revealed primarily by the sound of combustion and OH chemiluminescence. Proper orthogonal decomposition (POD), a statistical method, was employed on high-speed OH chemiluminescence data to study the flame dynamics. In addition to the mean field, information about the most energetic modes of flame dynamics can be acquired by employing this method [28]. POD is a procedure to decompose a given ensemble of data G (two-dimensional arrays in the present case) into a set of eigenfunctions. The details of this statistical method are not described here, but the outcome of this procedure is a certain number of orthogonal spatial eigenfunctions or modes $\Psi_i(x)$ and time coefficient $e_i(t)$ so that any snapshot *G(x,t)* taken at a time *(t)* can be reconstructed with a certain number of modes M according to Eq. (1).

$$G^N(x,t) = e_0\Psi_0(x) + \sum_{i=1}^{M} e_i\Psi_i(x) \tag{1}$$

Here, Ψ_0 or mode 0 represents the mean field. Subsequent modes represent fluctuation and thus the dynamics. In this method, modes are a linear combination of the snapshots, and the time coefficients can be computed projecting the modes onto snapshots according to Eq. (2)

$$e_j(t) = \Psi_j(x) \times G(x,t) \tag{2}$$

For further details on the theoretical background and on the implementation, the reader is referred to Refs. [29–32].

Proper orthogonal decomposition analysis revealed significant influences on the flame dynamics for changing the load and fuel, but not for adding pilot or central fuel. The energy content distributions of different modes at three conditions: (a) part load (NG) case, (b) base load (NG) case, and (c) base load (H_2) case are shown in Fig. 14 when neither pilot nor central fuel was added. Two dominant modes (first and second) contain about 29%, 30%, and 23% of the total fluctuating variance for the part load (NG), base load (NG), and base load (H_2) cases, respectively. Figure 15 presents modes 0–2 for the three cases, where mode 0 is representing the mean field while subsequent modes represent fluctuations to be added on top of it. The energy content of modes 1 and 2 is almost one order of magnitude larger than the energy of subsequent modes. The scale and sign of the contour plots are arbitrary as the modes are to be multiplied by the time coefficient. Mode 1 contains the axial fluctuation, while mode 2 indicates the rotational movement for the part load (NG) case. For the base load (NG) case, both modes contain axial and rotational movement. Finally, for the base load (H_2) case, mode1 represents axial fluctuation while mode 2 presents rotational movement.

As the time resolution of acquired data is high enough to capture the evolution of the time coefficients, FFT was performed on time coefficients, $e_1(t)$ and $e_2(t)$, of mode 1 and 2 for the three cases to reveal the frequencies of the modes' oscillations. For every case, both modes were found to oscillate with the same frequencies and therefore, only the frequency spectrum of mode 1 from each of the cases is considered for discussion. The frequency spectrum is plotted in Fig. 16 where frequency is expressed as Strouhal number (St), based on the reference velocity through the burner and the mixing tube diameter. In addition, the spectral

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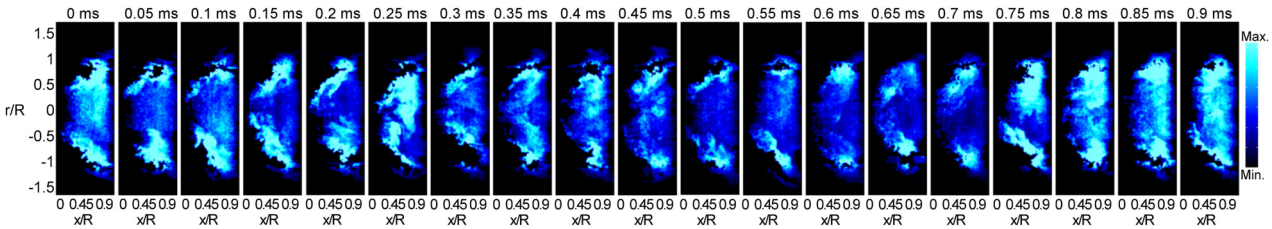

Fig. 10 The intensity fluctuation of the H_2 flame is shown by OH distribution in time sequence. Base load (H_2) case with PFR = 0 and CFR = 0.

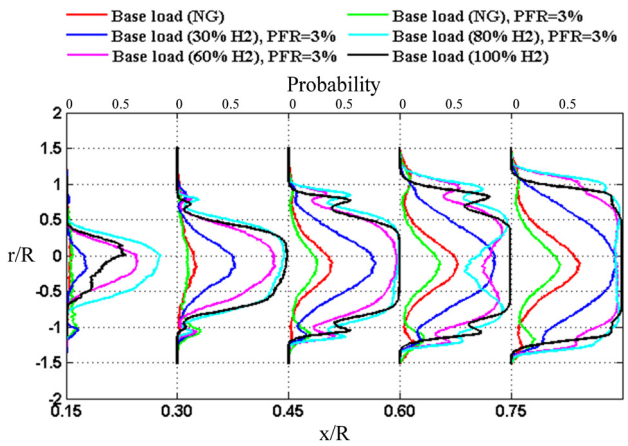

Fig. 11 Radial probability distributions of OH at five different axial positions obtained from OH-PLIF images for NG and H_2-enriched NG flame at base load

intensity is normalized by the maximum intensity value for the base load (NG) case.

For all cases, the frequency band is located at $0.1 < St < 0.2$. A previous study [22] in a similar burner showed the domination of acoustic oscillations in this band of frequencies, which was revealed by the high power spectrum density of the pressure fluctuations. Furthermore, the current study shows that the frequency of instabilities varies with the load and fuel changes. For the part load (NG) case, the axial mode dominates (cf. Fig. 15) over the rotational one and the spectral intensity is the maximum. For increasing the load, the rotational mode dominates, and the peak shifts toward a higher frequency with lower intensity. Combustion of H_2 further enhances the axial mode resulting in the highest frequency with the lowest intensity, which is consistent with the results found in the previous study [17]. The increase in frequency is related to the increase of local temperature which is attributed to higher heat release caused by the higher flame speed. With the temperature increase, the sound speed and by this, the eigenfrequencies of the combustion chamber increase.

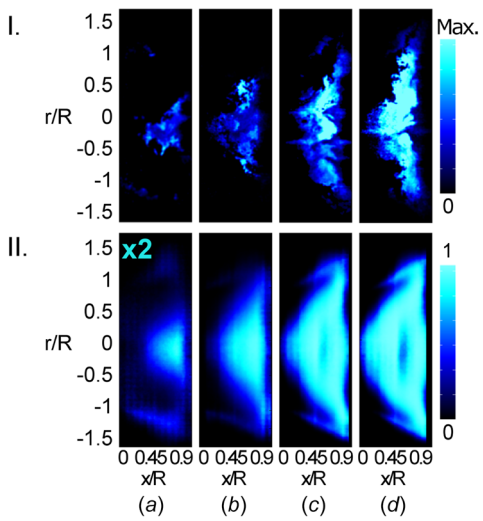

Fig. 13 OH-PLIF results of (*a*) NG/air, (*b*) 30 vol%, (*c*) 60 vol%, and (*d*) 80 vol% H_2 enriched NG flame at base load with PFR = 3%, CFR = 0 setting. I. Single-shots of OH-PLIF; II. PDMs of single-shots.

Emissions. The effects of load variations, pilot, and central fuel addition, and H_2 enrichment on NO_X production are shown in Fig. 17, even though it is not the main objective of the current study. The values for the NO_X are normalized by the value of $NO_{X(0)}$ at base load (NG) case. For H_2 enrichment of the fuel mixture from 0 to 100 vol %, an increase in NO_X production was observed, which is attributed to the increase in local flame temperature and the size of the postflame zone. By H_2 enrichment, the flame becomes more compact and intense due to the increase in flame speed and consequently, the local flame temperature and the size of the postflame zone increase. Higher temperatures and longer residence time in an extended hot postflame zone promote formation of thermal NO_X. Similar results were also observed previously [17]. For all the base load cases using NG and H_2, it may be seen that the upstream flame position increases NO_X, either it is

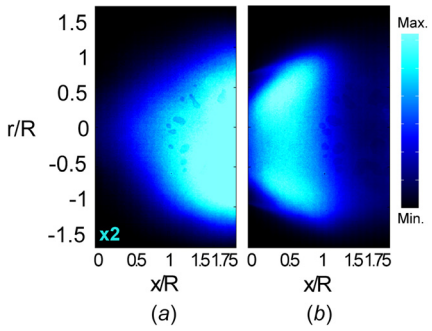

Fig. 12 Time-averaged OH chemiluminescence of (*a*) NG/air and (*b*) H_2/air flame at base load with PFR = 0 and CFR = 0

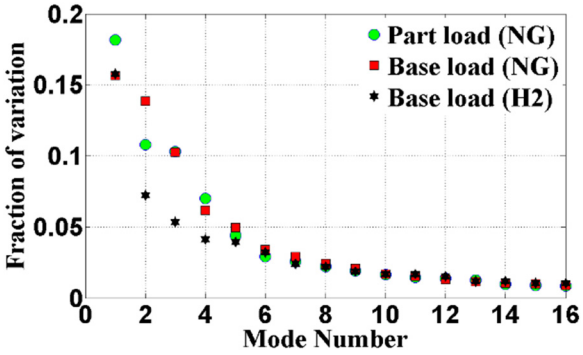

Fig. 14 POD variance distributions of modes 1–16

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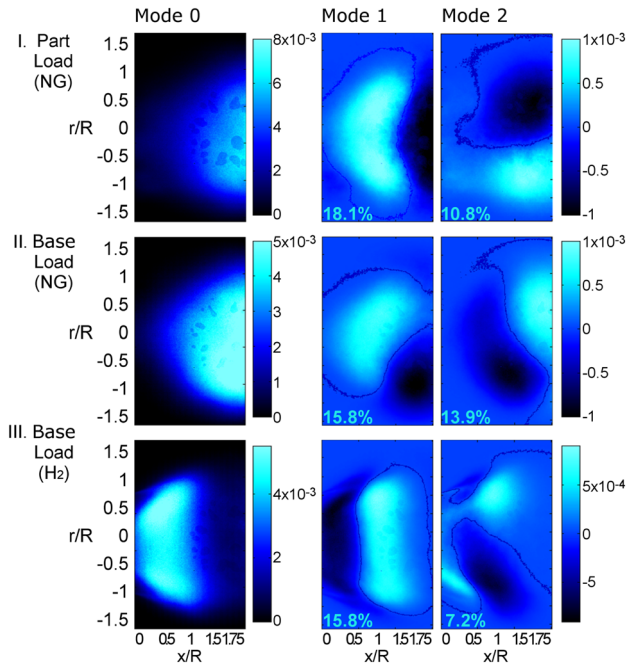

Fig. 15 Mode 0–2 are shown for three cases: I. Part load (NG), II. base load (NG), and III. base load (H_2) with contour plot of the intensity signal

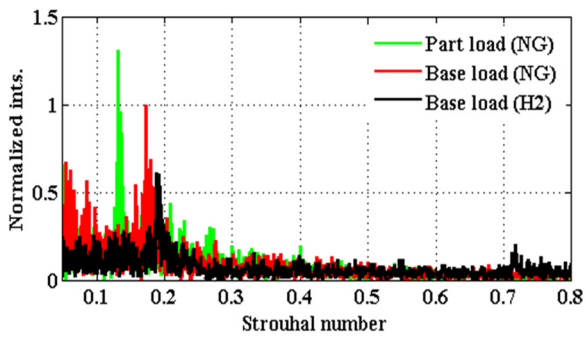

Fig. 16 Intensity versus Strouhal number for three cases: I. part load (NG), II. base load (NG), and III. base load (H_2)

due to CFR or H_2 content. In addition, for the part load (NG) condition, NO_X increased somewhat compared to the base load (NG) using the same PFR and CFR setting that may be due to the slightly higher residence time. In an engine, the reduced pressure

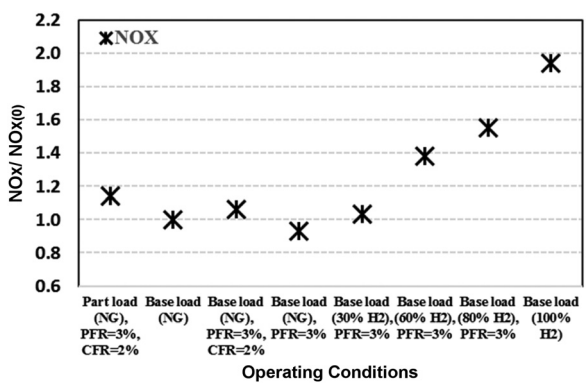

Fig. 17 Relative change in NO_X at different operating conditions

for part load acts to reduce NO_X, which is not captured in an atmospheric rig.

Conclusion

The effects of load and fuel variations on flame structure were investigated in a Siemens standard third generation DLE burner under atmospheric pressure conditions. High-speed kHz OH-PLIF and OH chemiluminescence imaging were employed to study the flame. Temporally resolved investigations of the flame structure revealed insights regarding the flame stabilization and dynamics.

The main flame was observed to be stabilized creating a CRZ. Addition of pilot fuel affected the ORZ significantly by strengthening the reaction process and therefore, the flame was found to be attached to the burner tip. When fuel was added at the central injector, the flame moved upstream without affecting the ORZ. For changing the load from part to base, the CRZ became stronger and more compact. The flame was stabilized close to the burner exit.

The higher reactivity and higher flame speed associated with H_2 affect the flame structure significantly. For H_2 enrichment in NG or replacing pure NG fuel with pure H_2, the CRZ that anchored the flame moved quite close to the burner exit. Without pilot fuel, even though the CRZ and ORZ became quite intense, the flame was not stabilized attaching to burner tip, whereas this attachment of the flame was observed for the pilot fuel addition.

The flame dynamics were affected by the variation of load and fuel type but not by the addition of pilot and central fuel. Application of the proper orthogonal decomposition reveals that the frequency of instability shifts toward higher values with decreasing spectral intensity for increasing the load part to base using NG and substituting fuel from NG to H_2.

In addition, an increase in NO_X was observed with enrichment of H_2 in the fuel mixture while a small change in NO_X was found for addition of pilot or central fuel as well as between part and base load.

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Nomenclature

AFR = air-to-fuel ratio
CFR = central fuel to total fuel ratio
CMOS = complementary metal oxide semiconductor
CRZ = central recirculation zone
DLE = dry low emission
f = focal length
FFT = fast Fourier transform
H_2 = hydrogen
HC = hydrocarbon
ISL = inner shear layer

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K = kelvin
LBO = lean blowout
MW = megawatts
NG = natural gas
NO_X = NO and NO_2 (nitric oxide and nitrogen dioxide)
OH = hydroxyl
ORZ = outer recirculation zone
OSL = outer shear layer
PDM = probability density map
PFR = pilot fuel to total fuel ratio
PLIF = planar laser-induced fluorescence
POD = proper orthogonal decomposition
R = burner exit inner radius
SGT = Siemens gas turbine
St = Strouhal number
UV = ultraviolet
ϕ = equivalence ratio

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